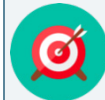
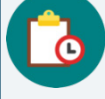


10.3 Radian Measure

Lesson Introduction

 Benchmark	MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.		 Learning Targets	- I can define radian. - I can explain the relationship between the length of the arc intercepted by an angle to the radius.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is a radian? - What are radians used for? - What is the relationship between the length of the arc intercepted by an angle and the radius of the circle?			
 Lesson Narrative	The focus of this lesson is for students to be able to define radian and explain the relationship between the radius, central angle, and arc length. Students will be looking at the circumference and arc length in circles of various radii to make discoveries about this relationship. They will be able to define radian as a ratio of arc length to radius.			
 Component Summary	10.3.1 Warm-Up (~5 min.)	Choosing between two bank deposit scenarios (<i>SE p. 130</i>)	10.3.3 Exploration (10-15 min.)	Discovering radians in a whole circle (<i>SE p. 132</i>)
	10.3.2 Exploration (10 min.)	Discovering radian measure (<i>SE p. 131</i>)	10.3.4 Bring It Together (10-15 min.)	Connecting arc length to radius (<i>SE p. 133</i>)
	10.3.5 Wrap-Up (~5 min.)	Reflection on the learning of the lesson (<i>SE p. 134</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Radian <u>Additional Glossary:</u> - Arc Length		N/A	10.3.2 Exploration contains a Desmos activity to accompany the question prompts.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may harbor the misconception that a radian measure must be a multiple of π. - Students may have difficulty simplifying fractions to see the connection between the ratios of the circumference to the given radii. - Students may use the area formula during 10.3.3 Exploration instead of the circumference formula.	- Consider having large posters with differently sized circles drawn on them, and have students measure the radius and arc length using a ruler and a string. - Students will not be converting from degrees to radians yet.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect This lesson provides students the opportunity to discover radians. In 10.3.2 Exploration, encourage students to compare and connect the central angle, arc length, and radius. Allow students to discuss with other students to strengthen their understanding.	MA.K12.MTR.5.1: Patterns and Structure Students will be discovering the connection between the radius, central angle, and arc length. Use this as an opportunity for students to explore the patterns and structures to be able to conjecture the relationships.
 Differentiate Instruction	10.3.2 Exploration provides both visual and kinesthetic entry points for learners to engage with the content. The tables in this exploration provide additional support for visual learners to see the connection between the radius, central angle, and arc length. Engaging the class in discussions by having students share observations they made and explain their discoveries will help auditory learners deepen their understanding.	
Lesson Notes:		

10.3.1 Warm-Up: Would You Rather?

Component Overview: Students will identify which option they would rather have. Students will be asked to explain their choice using mathematics.



10.3.1 Warm-Up: Would You Rather?

Would you Rather...		
Deposit \$5 in the bank and have it triple each week for 4 weeks?	OR	Deposit \$3 in the bank and have it quadruple each week for 4 weeks?

- Which would you rather choose? Justify your reasoning using mathematics.

Answers will vary. Putting \$5 in the bank to triple for 4 weeks would give me \$405, whereas putting \$3 in the bank and quadrupling it for 4 weeks would give me \$768. So, I would rather deposit \$3 in the bank and have it quadruple each week for 4 weeks.



Do you think the two options will ever equal the same amount of money? Why or why not?

10.3.2 Exploration: Discovering Radian Measure

Component Overview: In this Exploration, students will work with Desmos to explore the impact the radius measure has on the arc length and the central angle. Students will be able to conclude that radians are based on the number of radii wrapped around the circumference of a circle, and the radian measure of a central angle is equal to the length of the arc intercepted by the central angle divided by the radius of the circle.



10.3.2 Exploration: Discovering Radian Measure

Angles that are measured in degrees can also be measured in **radians**. Measuring in radians is similar to measuring distance using other units, such as inches or feet. Radians can be written as a decimal or in pi (π) form.

With your partner, work through the Desmos activity "Discover Radian Measure." Complete the steps to answer each question.

Take a moment to look over the activity and the information provided.

- Slider " r " adjusts the radius of the circle. The radii for the activity range from 0 to 3, increasing by 0.5.
- Slider " c " adjusts the central angle.
- The central angle, $C_{entralAngle}$, is measured in radians, written as a decimal. Radians for the activity increase by 0.25.

Drag slider r to the right and left to observe how the measurements of radius, central angle, and arc length change as the r value changes. These values are in rows 2, 5, and 6 respectively of the activity.

Hit play on slider c to watch how the measurements of radius, central angle, and arc length changes. These values are in rows 2, 5, and 6 respectively.

1. Adjust slider c , $C_{entralAngle}$, so that the arc length, $A_{rcLength}$, equals the radius, r . Complete the table for three different measures of r .

Answers will vary.

Radius (r)			
$C_{entralAngle}(c)$			
$A_{rcLength}$			



The 10.3.2 Exploration can be done in pairs, in groups, or as a whole class. If technology is limited, consider projecting the activity for the class and play the sliders. Make the adjustments as prompted by the activity.



Consider splitting students into groups, with some groups exploring the impact of radius measure on arc length, and the other groups exploring the impact of angle measure on the arc length. Have students come back together and encourage students to explain their findings and think about the variation among arc length, central angle, and radius.

10.3.2 Exploration: Discovering Radian Measure (continued)

2. Describe the relationship observed in the table between the measure of the arc length and the measure of the radius.

The measure of the arc length is only equal to the radius if the central angle is measured to be 1 radian. Arc length equals the central angle multiplied by the radius.

3. Adjust slider r , radius, so that the arc length, $A_{rclength}$, is twice the radius, r . Complete the table for three different measures of r .

Answers may vary.

Radius (r)			
CentralAngle (c)			
$A_{rclength}$			

4. Describe the relationship observed in the table between the measure of the arc length and the measure of the radius.

Answers may vary. The arc length is only double the radius if the central angle measures 2 radians. When the central angle measures 2 radians, the arc length will be double the radius.



Consider asking the following:

- How does the radius vary with the arc length?
- How does the radius vary with the central angle?
- Is the relationship direct or inverse variation?

10.3.2 Exploration: Discovering Radian Measure (continued)

5. Complete the statements based on your work in questions 1-4.

A radian is a unit used to express the measure of an

angle. Radians are based on the number of ☐ angles
☐ radii

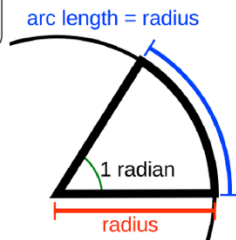
of a circle being wrapped along the ☐ diameter
☐ circumference

of the circle. The radian measure of a central angle is

equal to the length of the ☐ arc
☐ radius

intercepted by the central angle divided by the

☐ radius.
☐ arc length.



To help students understand the relationship between the radius and the arc length, consider having students do a hands-on activity:

- Draw a circle with a piece of string and a pencil.
- Place the string on the circumference of the circle.
- Use the end points on the string to create a central angle.

Ask students, “What do you notice about the arc length? How is it related to the radius?”

6. Describe how the diagram shows the relationship between the radius and arc length.

Answers will vary. It shows that the central angle (in radians) measure equals the $\frac{\text{arc length}}{\text{radius}}$. The ratio of the arc length to the radius will be equal to the central angle (measured in radians).

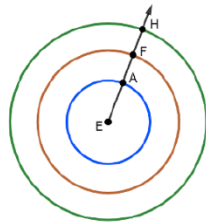
10.3.3 Exploration: Discovering Radians in a Whole Circle

Component Overview: Using different radii, students will discover that the ratio of the circumference to the radius will equal 2π , which is the radian measure of a whole circle.



10.3.3 Exploration: Discovering Radians in a Whole Circle

1. Consider the diagram depicting three concentric circles each with center E . Point A lies on the blue Circle E , and has radius $EA = 1$ unit. Point F lies on the brown Circle E , and has radius $EF = 2$ units. Point H lies on the green Circle E , and has radius $EH = 3$ units.



- a. Complete the table with your partner.

Circle	Circumference	Ratio of Circumference to Radius
Radius $EA = 1$ unit	2π	$\frac{2\pi}{1} = 2\pi$
Radius $EF = 2$ units	4π	$\frac{4\pi}{2} = 2\pi$
Radius $EH = 3$ units	6π	$\frac{6\pi}{3} = 2\pi$

- b. Discuss any observations you made about the values in the table with your partner. Summarize your discussion.
The ratio of the circumference to the radius of each Circle E color is equal.
- c. Write a conjecture about the ratio of the circumference to radius and a whole circle measured in radians.
The ratio of the circumference to the radius of any whole circle will be equal to 2π , which is the radian measure of any whole circle.



Consider asking the following:

- Do you think the ratio of the circumference to the radius will always be 2π ?
- If 2π is the radian measure of a whole circle, what do you think would be the radian measure of half a circle? What about a quarter of a circle?

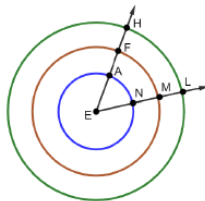
10.3.4 Bring It Together: Relating Arc Length to the Radius

Component Overview: Students will use concentric circles to conclude the proportional relationship among arc length and radius.



10.3.4 Bring It Together: Relating Arc Length to the Radius

- Concentric circles with center E are shown. Points A and N lay on the blue Circle E , and has radius $EA = 1$ unit. Points F and M lay on the brown Circle E , and has radius $EF = 2$ units. Points H and L lay on the green Circle E , and has radius $EH = 3$ units.



Describe how the arclengths AN , FM and HL are related.

Answer will vary.

The arc lengths are proportional.

For instance, $\frac{AN}{1} = \frac{FM}{2} = \frac{HL}{3}$.

Which is the same as: $AN = AN, FM = 2AN, HL = 3AN$

- Use words from the bank below to complete the statements.

angle	arc	radian	radius
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The length of the **arc** intercepted by a(n) **angle** is proportional to the **radius** of the circle. The **radian** measure of the angle is the constant of proportionality.



$$\text{Radians} = \frac{\text{Circumference}}{\text{Radius}} = \frac{2\pi r}{r} = \frac{2\pi}{1}$$

$$360^\circ = 2\pi \text{ radians}$$

$$180^\circ = \pi \text{ radians}$$



Notice and Wonder

Consider providing students with the opportunity to write down things they notice and wonder about the concentric circles. This will allow students the chance to make observations before answering the question.

10.3.5 Wrap-Up: Cool Down

Component Overview: Students will summarize what they learned from the lesson by writing a note to help an absent student understand the relationship between the length of an arc intercepted by an angle and the radius.



10.3.5 Wrap-Up: Cool Down

1. Your friend was absent today and will need notes from class. Write a quick note to help them understand relationship between the length of an arc intercepted by an angle and the radius.

Answers will vary.

The arc length intercepted by an angle is proportional to the radius of the circle.

The radius multiplied by the central angle (measured in radians) will equal the arc length.



Consider having students reflect and document the challenges they faced in this lesson as well as what they learned. They can use their notes to prepare for future lessons.